

The empirical recurrence for OEIS A253980: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A253980 records “Empirical recurrence of order 68”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 178$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 68, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 68 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A253980 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with every 3×3 subblock diagonal maximum plus antidiagonal maximum nondecreasing horizontally and vertically.”. It has offset 1 and begins

21304, 417518, 8727838, 179261149, 3672381306, 75620185999, 1556452499274, 32027982651882, ...

The entry states:

Empirical recurrence of order 68 (see link above).

As of the “Last modified” line on the live entry (Jan 28 2025 15:59:22, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1748$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 178$ of the 1748, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 178$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 356$ exact terms of the model returns order 68 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	28	c_{18}	32699882935471	c_{35}	21088569848554387200	c_{52}	−275833828457276
c_2	−145	c_{19}	−80824418372722	c_{36}	−6175948085185980672	c_{53}	−473425081035810
c_3	−290	c_{20}	56284941969440	c_{37}	−35119055493109299200	c_{54}	1186363267494275
c_4	3956	c_{21}	−46956907765996	c_{38}	87287433883317381120	c_{55}	−1092181383697229
c_5	−34092	c_{22}	−1251829135370140	c_{39}	−71152746818372299776	c_{56}	3550825316851541
c_6	509816	c_{23}	7345279239088264	c_{40}	−1193184180774223872	c_{57}	2164984478479417
c_7	−9183232	c_{24}	−15642954340140224	c_{41}	79838548171995246592	c_{58}	−597837456528830
c_8	47960426	c_{25}	34991707665592528	c_{42}	−145197454741129986048	c_{59}	5526564159110051
c_9	37779032	c_{26}	−30230892226810528	c_{43}	98219181166194999296	c_{60}	−175101205924628
c_{10}	−678709222	c_{27}	−167431433226489088	c_{44}	46278324649743876096	c_{61}	−382039107756946
c_{11}	2723203204	c_{28}	525110680374564432	c_{45}	−234303966570154885120	c_{62}	1100951823081642
c_{12}	−16659134876	c_{29}	−858915930555997632	c_{46}	414271824050006786048	c_{63}	−101657551517087
c_{13}	53539215576	c_{30}	942642794728024464	c_{47}	−338898926865739415552	c_{64}	3011881646655170
c_{14}	−113330978436	c_{31}	−581448516624828640	c_{48}	28683437128719269888	c_{65}	269960011335283
c_{15}	67284010856	c_{32}	−1137297384843599680	c_{49}	472882511558017548288	c_{66}	−46938415272548
c_{16}	1656306886479	c_{33}	9956885340665483712	c_{50}	−1029184145759225774080	c_{67}	410390516044136
c_{17}	−8030982358532	c_{34}	−22016001838561733760	c_{51}	928564152237266305024	c_{68}	−12311715481324

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 68, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $68 + 4$ terms removed returns order 68 as well, so $\deg q_{\min} = 68 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $68 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 68 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A253980.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.